

# Deloitte Data Analytics Job Simulation Reflection

Completing the Data Analytics Job Simulation offered by Deloitte Australia on Forage has given me a broader view of how data analytics is used to solve business problems. Having chosen to study Business Analytics and Information Technology as my major, I have become aware of the role of data in decision-making. Nevertheless, this simulation has allowed me to realize this notion more practically, since it was not an abstract exercise dealing only with theoretical knowledge and technical skills. Instead, I was required to analyze data in a business context, apply different tools, and produce results that could help a client better understand a business issue. The simulation included two main tasks: analyzing machine telemetry data in Tableau and analyzing gender pay equality data in Excel.

The first task involved machine telemetry data collected from Daikibo factories in Japan, Germany, and China. The company collected data from a variety of machines every ten minutes, and the objective was to find out which factory experienced the highest level of machine downtime and which device type contributed most to that downtime. First, I imported the JSON dataset into Tableau. After this, I created a calculated field called “Unhealthy,” assigning a value of 10 whenever a machine was unhealthy, since each record represented a ten-minute interval. Following this, I created two bar graphs: the first graph showed downtime per factory, while the second showed downtime per device type.

This was probably one of the most exciting tasks in the simulation since it allowed me to see how data can be converted from raw data to something more comprehensible. If I had been scanning the huge JSON file directly, it would have been rather hard for me to see which factory experienced the most issues. However, after converting the data into visual form, it became rather easy to see the answer to my question. Daikibo Factory Seiko location had the biggest downtime, and by filtering that particular factory in the dashboard, I could find out what kind of machines caused the most downtime. I found the interactive filter very intriguing since it allowed me to see that data visualization can serve several purposes, not just make data more presentable in graphs.

Before undertaking this task, my understanding of data visualization was basically limited to something that happened after analysis had been performed. Through the exercise, I have gained knowledge that data visualization may also be an integral part of the analysis process. The ability to organize data visually and filter it by location makes it easier to move from a general question to a more specific one. This is especially useful in a business context because managers do not need to look through every individual data point. Instead, they need clear information that helps them understand where a problem exists and what deserves further attention.

The second task was related to the application of forensic technology in the context of gender pay equality. Daikibo had reported some internal grievances regarding potential gender

disparities in their salary structure, and an equality score had already been developed based on various jobs and factories. My assignment was to categorize them into one of the three classes, i.e., Fair, Unfair, and Highly Discriminative. The spreadsheet contained information regarding the factory name, job, and equality score. I added a fourth column for the classification.

Rather than manually entering a category into every row, I used an Excel formula based on the absolute value of the Equality Score. Scores between -10 and +10 were classified as Fair, scores outside that range but within -20 and +20 were classified as Unfair, and scores below -20 or above +20 were classified as Highly Discriminative. I chose to use a formula because it made the process more efficient and reduced the possibility of inconsistent classification. When I compared my results with the model answer provided by Forage, I noticed that the example answer showed the final text classifications directly instead of formulas. However, my results matched the intended categories. This showed me that there can be different ways to complete the same analytical task as long as the logic and final results are correct.

Another interesting part of this project was how data analysis was applied in a totally different way. While the first one was about operational performance and machine downtime, the second was about employee remuneration and justice. The tools used were also different: Tableau proved to be more helpful in the case of large datasets exploration and visualization, whereas Excel sufficed for smaller classification tasks. It made me remember that data analysis does not always require the most advanced tool; it depends on the situation and the task at hand: sometimes you need a dashboard, and sometimes an Excel formula will do just fine.

One challenge I encountered during the simulation was learning how to use Tableau correctly. Unlike Excel, Tableau seemed to me less clear, which is why I needed to be more attentive when working with the import of a JSON file, the creation of the calculated field, creating two visualizations, and the application of the filter on the dashboard. But the provided instruction helped me realize how everything needs to be done. After the completion of the dashboard, I checked it against the professional example answer and understood that everything worked as it should. If I have an opportunity to work on such a task again, I would like to learn more about Tableau beyond the required steps so that I can use more advanced visualization features and create more customized dashboards.

Overall, this simulation helped me better understand what data analytics can look like in a professional environment. One of the most important lessons I learned is that technical skills are only one part of the process. The main purpose of analysis is to turn data into information that helps someone understand a problem and make a better decision. In the first task, this meant identifying which factory experienced the most downtime and which device type contributed most to that downtime. In the second task, it meant organizing pay equality scores into categories that could help identify areas that might require further investigation.

As a Business Analytics and Information Technology major, this project was closely related to my academic interests. I have had an opportunity to practice in using Tableau, improve my spreadsheet skills and apply the process of data analysis in answering actual business questions. The main lesson that I have learned during this experience is that analytics is not only about calculating but also about understanding the sense of calculations made. It would be great for me to engage in similar assignments in the future.